

Urinalysis

Today's Materials

- Gloves
- Urine collection cups
- Disposable transfer pipettes
- Microcentrifuge tubes
- Centrifuge
- Urinalysis test strips
- pH 5-9 range test strips
- Urinometers (for specific gravity)
- Sedi-stain dye
- Slides and cover slips
- Patients A-E & 1-3 Urine Samples
- Disposal containers for liquid, contaminated slides, and cups
- Blank Pages in Lab Notebook
- Extra Textbooks/Manuals in Room

Notes:

- Wear gloves throughout the experiments and dispose of all materials in designated areas.

Activity 1: Analysis of Urinary Sediments

(Adapted from Sierra College Bio 6 Lab Manual)

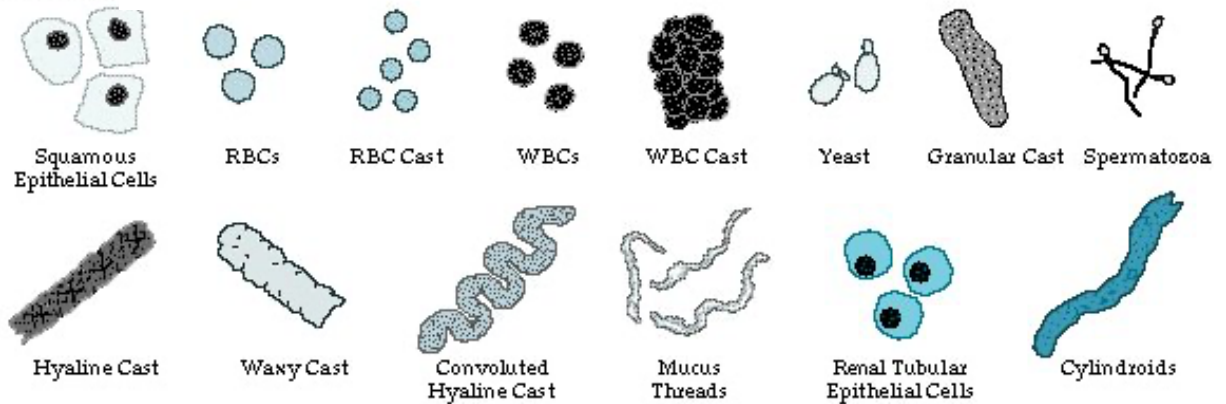
Introductory Notes

- **Color:** Fresh urine is typically clear/ pale yellow to amber. The yellow color is due to **urochrome (urobilin)**, a pigment metabolite from the destruction of hemoglobin. In general, color varies from pale to deep amber based on the solute concentration in urine. The more concentrated the darker the color, and the less concentrated (dilute) the lighter the color. Abnormal color may also occur when certain foods are consumed such as beets. Some medications, bile, and blood may also alter the color.
- **Turbidity:** Healthy fresh urine is clear to slightly cloudy, while abnormal is cloudy or opaque. Excess *turbidity* results from the presence of suspended particles in the urine. Common causes of abnormal turbidity include: numerous crystals, bacteria, lipiduria, and fecal contamination.
- **Odor:** Fresh urine is slightly aromatic, but bacteria may give an ammonia- like odor when left standing. Some drugs, vegetables (such as asparagus), and diseases (such as diabetes mellitus) alter the odor.
- **Normal urine constituents:**
 - Water
 - Urea
 - Sodium
 - Potassium

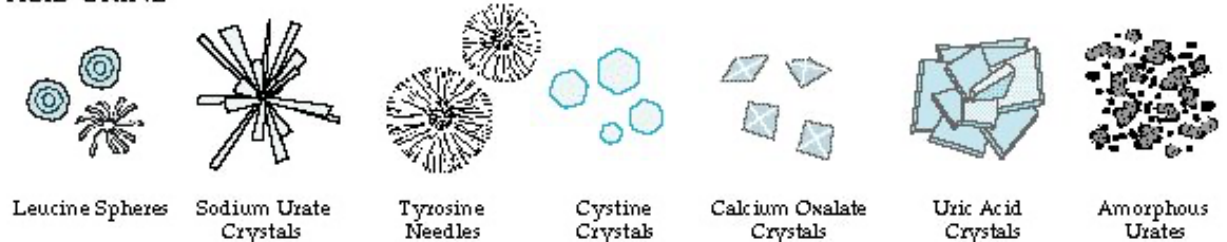
- Phosphate
- Sulfate ions
- Creatinine
- Uric acid
- Bicarbonate ions
- **Test strip constituents:**
 - **Leukocytes:** White blood cells or pus in urine is a condition called, *pyuria*. It indicates inflammation of the urinary system.
 - **Nitrites:** Nitrites in urine might indicate a bacterial infection, particularly *E.coli* or other gram negative rods.
 - **Urobilinogen:** Urobilinogen is formed in the intestines by bacterial action on bilirubin. Some urobilinogen is reabsorbed into the blood and either excreted back into the intestine by the liver or excreted by the kidneys in the urine. Absence of urobilinogen may indicate renal disease, while increased levels may indicate hepatitis A or cirrhosis.
 - **Protein:** Protein (or albumin) in urine is a condition called, *albuminuria*. Albumin is an important plasma protein that plays a major role in maintaining osmotic pressure and is too large to pass through the glomerular filtration membrane. Presence of albumin in urine indicates abnormally increased permeability of the glomerular filtration membrane.
 - **pH:** The pH of urine ranges from 4.5- 8.0, but the average is 6.0 (slightly acidic). Diet is the main factor to affect the pH. For example, a diet high in protein and whole wheat increases the acidity in urine, on the other hand, a vegetarian diet decreases the acidity.
 - **Red blood cells:** Red blood cells in urine is a condition called, *hematuria* and is an indication of pathology in the urinary tract. RBC's are too large to pass through the glomerular filtration membrane and should not be present in urine. Possible causes include urinary tract infection, kidney stones, or physical trauma to urinary organs. Menstruating females may also experience accidental contamination of the urine sample with the menstrual flow.
 - **Specific gravity:** The relative weight of a specific volume of liquid compared with an equal volume of distilled water. Specific gravity for distilled water is 1.000, while urine ranges from 1.001- 1.030 depending on solute concentration. Conditions that produce urine with high specific gravity include limited fluid intake, fever, and kidney inflammation. Excessively concentrated urine may lead to substances in the kidney to precipitate or crystalize, forming *kidney stones*.
 - **Ketone bodies:** Excessive amounts is referred to as *ketonuria*, and indicates abnormal metabolic processing. This is expected during starvation, or very low carb diets when the body begins to rely on lipid for energy production.
 - **Bilirubin:** Bilirubin (bile pigment) in urine is a condition called, *bilirubinuria*. This usually indicates liver pathology, such as hepatitis, cirrhosis, or bile duct blockage.

- **Glucose:** Glucose in urine is a condition called, *glucosuria* and is an indication of high blood sugar levels. Normal blood glucose level is between 80 and 100mg/100ml of blood.
- **Urine sediments:** The microscopic examination of the urine sediment is a big part of urinalysis. Sediment presence is normal and necessary for the proper interpretation of results. Below are examples of sediments found in different urine pH levels. **Casts** are formed in renal tubules and may be due to concentrated urine or renal pathology. **Crystals** may form in non-freshly voided urine, acidic urine, or alkaline urine.

NORMAL pH



ACID URINE



ALKALINE URINE



Procedure for Analysis of Sediments

Wear gloves throughout the experiments and dispose of all materials in designated areas.

At the start of lab, you will go to the bathroom and collect a urine sample in the provided specimen container.

Part 1- Preparing a Centrifuge Tube & Recording General Observations

- 1- Place 2ml of your urine sample into a centrifuge tube. Label your tube with your initials.
- 2- Place the tube in the centrifuge, noting the position of your sample. In order to keep the centrifuge balanced, place tubes opposite of each other. Your instructor will set the centrifuge to the highest speed for 5 minutes.
- 3- While you wait for the centrifuge to complete the spin cycle, fill in your observations regarding your urine in the Part 1 datasheet on the next page.
- 4- At the end of 5 minutes, turn off the centrifuge, wait for it to stop spinning and remove your tube.

Part 2- Making a Microscopic Slide

- 1- Obtain your tube from the centrifuge.
- 2- Carefully pour the fluid back into your collection cup without disturbing the sediment in the bottom of the centrifuge tube.
- 3- Using the pipette, remove a drop of sediment from the tube, place it on the slide, add one drop of sedi-stain (dye), and cover it with a cover slip.
- 4- Observe the material on the slide, then draw and label them on Part 2 of the datasheet on the following pages. You will probably find various epithelial cells, casts (hardened cells, proteins, fats, etc.), crystals and mucous threads.

Part 3- Test for Constituents

- 1- Obtain a reagent (test) strip to detect the presence of abnormal constituents in your urine.
- 2- To use a reagent strip, dip it into the urine, remove excess urine against the side of your cup, and compare the color of the reagent strip to the color scale on the side of the jar.
- 3- Be sure to carefully read the instructions on the side of the jar before using the strip. Notice that all tests must be read within a certain time after dipping into the urine. List the constituents that the strip detects on Part 3 of the datasheet.
- 4- When done, dispose of the strips in the designated container.

Disposal instructions:

- Pour excess urine into the toilet.
- Dispose of the urine cups, reagent strips, pH paper, centrifuge tube, slide and cover slip in the biohazard bag.

URINALYSIS- DATA SHEET**Part 1: Answer the following questions.**

- 1) Record the color of your urine. _____
- 2) What influences the yellow color of urine? _____
- 3) Record the odor of your urine. _____
- 4) What most influences the odor of urine?

- 5) Why will your urine vary in color throughout the day?

- 6) Is your urine clear or turbid? _____
- 7) What causes turbid urine? _____
- 8) Why does the pH of urine vary? _____
- 9) Give several reasons why specific gravity of urine varies throughout the day.

Part 2

Draw and label your sediments below.

Part 3: Write your tested values for your urine samples in the table below.

Urine Characteristic	Normal Value	Tested Value Control Urine
Color	Variable	
Odor	Variable	
Turbidity	Variable	
Leukocytes	Negative	
Nitrites	Negative	
Urobilinogen	Positive	
Protein	Negative	
pH	4.5-8.0	
Red Blood Cells	Negative	
Specific Gravity	1.001-1.030	
Ketone Bodies	Negative	
Bilirubin	Negative	
Glucose	Negative	

Activity 2: Abnormal Urine Sample Analysis

For each patient's urine you will be measuring the specific gravity, blood/hemoglobin, leukocytes, protein, ketones, glucose and pH. Record your data in the Table provided.

- 1) Obtain 5 test tubes and label A, B, C, D, E respectively.
- 2) Add the appropriate artificial urine sample to each tube. The amount to add will be announced in class.
- 3) Use pH paper to measure the pH.
- 4) To test for all other substances use the Chemstix.
Test one urine sample at a time.
Dip the test stick into the sample, remove and place on the paper towel for the appropriate amount of time. (The time of when to read the results is listed on the Chemstix canister)
Record all your results from this one patient before moving onto the next.
Hold the stick against the canister to read.
- 5) When your experiment is finished, empty the test tubes in the container provided and place the test tubes in the appropriate container. Dispose of all trash. Wipe your table down.
- 6) Once you have cleaned up complete your Table with observations on the possible condition of each patient. Follow any instructions given in class about how to complete the Table.

There is one 'control'/normal patient, and 4 patients with one major abnormality in their urine.

	#A	#B	#C	#D	#E
COLOR					
TURBIDITY					
pH					
GLUCOSE					
PROTEIN					
BLOOD					
KETONES					
LEUKOCYTES					
SPECIFIC GRAVITY					
DIAGNOSIS?					

- 7) Use the urinometers to test the specific gravity of urine samples 1-3 which are set up at the front of the room.
Measure specific gravity using the urinometer. Handle with care.
Learn how to read the calibration on the urinometer. Record the value where the bottom of the meniscus crosses the stem.

Urine 1:
Urine 2:
Urine 3:
Which urine is most dilute?
Which urine is most concentrated?

Diseases which influence urine volume – There are some diseases that affect how much urine we make every day. Some will cause us to drink too much water and urinate more often and more volume. Some will cause us to make much less urine.

Do a little research: Explain only how urine **volume** and **concentration** may change in each of the following diseases when the disease is untreated.

Explain the mechanism in each case.

- i. Diabetes Mellitus:
- ii. Nephrogenic Diabetes Insipidus (note diabetes insipidus is VERY different than diabetes mellitus – do some reading before answering this question):
- iii. Kidney Failure: There is a big difference between early kidney failure and late/end stage kidney failure – research what happens to the amount/concentration of urine we make with these two very different stages:

Activity 3: Filtration, Absorption, Secretion Activities

Refer to the instructions provided by your instructor for any additional experiments (if available).